Theory Assignment

Day-05

problem 1: The Steepent Ascent.

TaskA: heometrically, if you evalate the gradient vector at a specific point on a 3D surface, what specific direction does that vector point towards?

TaskB: In ML, our goal is to minimize a loss function $L(w)$. If the gradient points in the direction in TaskA mathmatially state the update rule we must use to adjust our weights w to iteratively reach the bottom of the valley.

TaskC: What role does the magnetude of the gradient vector $\|\nabla L\|$ play in this process? What happens to this magnatude as we get closer to the very bottom of the loss valley?

Solution:

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Task A: ~~The direction of~~

The gradient vector always point in the direction of steepest ascent. (the direction that increase the fn value the fastest.

Task B:

To minimize the loss, we must move the opposit direction of the gradient.

The update rule is:

$$W_{new} = W_{old} - \eta \left(\frac{\partial L}{\partial w}\right) \nabla L$$

Task C:

The magnitude $\|\nabla L\|$ represent steepness of the slope. As we get closer to the bottom of the loss valley, the magnitud of the gd shrinks and approaches 0.

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problem 2: Contour maps and orthogonality.

A contour map represents a 3D surface in 2D by drawing level curves.

Task A: If you walk exactly along a level curve, what is the rate of change of the function?

Solution: ... the f(n) dosen't change at all.

Task B: State the geometric relation Between the gradient vector evaluated at a point and the level curve passing through the exact same point.

Solution: 90°/orthogonal.

Task C: Using Concept of Directional Derivative ($D_u f = \nabla f . u$) mathmatially prove your answer to Task B.

Solution: If the direction u, follow the curves.

$D_u f = \nabla f . u = 0$.

if dot product of two non zero vector is 0, they must be orthogonals.

Problem 3: The Illusion of Zero.

In single variable calculus, setting the derivative to zero ($f'(x)=0$) helps you find the min of a curve.

In multivariable calculus, we use gradient is the 0. $\nabla f=0$. These are called critical points.

Task A: If $\nabla L(w)=0$ does this absolutely guarantee that you have to found the optimal weight for your ML models?

Solution: No. A gradient of 0 simply means the surface is perfectly flat at that instant. It could be min, max, or saddle point.

Task B: explain the concept of saddle point in multivariable landscape. Why is a saddle point considered a critical point, and why is it a massive problem for training deep Neural Networks?

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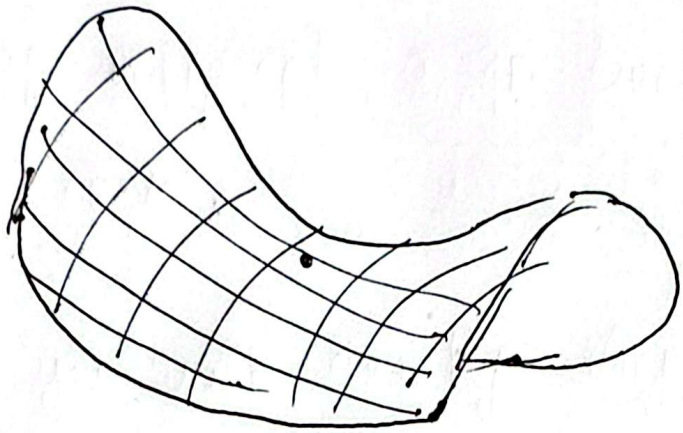
Solution:



3D. fig

A saddle points look like the saddle on a horse. it curves upward in one direction and downward in Another direction.

It is a critical problem in DNN Because the gradient Becomes 0,

Causing Gradient Descent algorithm to Stall or get Stuck before reaching the true global minimum.

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